Performance Analysis of the BATMAN Proactive Routing Protocol in a 4-Node Wireless Ad Hoc Network: NS2 Simulation and MAC-Layer Evaluation

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***Abstract*—This paper presents a detailed performance evaluation of the Better Approach To Mobile Ad hoc Networking (BATMAN) protocol using NS2 simulation on a 4-node linear wireless chain topology. We implement a Tcl-level BATMAN overlay that faithfully reproduces the core OGM (Originator Message) flooding, transmit-quality (TQ) computation with hop-penalty degradation, and routing table maintenance. The simulation runs alongside AODV as the underlying NS2 routing engine, allowing direct comparison of proactive (BATMAN) versus reactive (AODV) route discovery paradigms. Our results show that BATMAN achieves route convergence in 30 ms (3 $\times$ 10 ms RTR-level hop delay), providing a proactive advantage of 2.17 s before the first CBR data packet. MAC-layer analysis confirms steady-state end-to-end delays of approximately 9.62 ms with very low jitter ($<$1.5 ms deviation from the 200 ms CBR interval). The TQ decay follows the theoretical $\mathbf{TQ}_{n+1} = \mathbf{TQ}_n \times (255-30)/255$ model with perfect fidelity. We also measure AODV's on-demand route discovery time of 26.6 ms and quantify the MAC overhead attributable to IEEE 802.11 CSMA backoff and ACK exchange. All trace data, AWK analysis scripts, and NS2 Tcl source are provided for reproducibility.**

***Index Terms*—BATMAN, MANET, OGM flooding, route discovery time, NS2 simulation, MAC delay, TQ metric, proactive routing, AODV comparison.**

I. Introduction

Mobile Ad hoc Networks (MANETs) operate without fixed infrastructure, requiring nodes to cooperate in route maintenance. The choice between *proactive* protocols—which maintain routing tables continuously—and *reactive* protocols—which discover routes on demand—fundamentally shapes application latency and control-plane overhead.

BATMAN [1] is a proactive link-state-free protocol that avoids the distributed Dijkstra computation of OSPF and OLSR. Instead, each node periodically broadcasts an Originator Message (OGM) with a transmit-quality score. Neighbours rebroadcast received OGMs after applying a *hop penalty*, creating a distributed gradient that points every node toward the best next-hop for each originator. BATMAN was standardised as B.A.T.M.A.N. v1 and later evolved into batman-adv (layer-2 mesh).

In this work we evaluate the **v1 BATMAN** algorithm on a 4-node linear chain via NS2 [2], with the following contributions:

- A complete Tcl overlay implementing OGM flood, TQ computation, deduplication cache, and route-install logic, annotated for AWK post-processing.
- RTR-layer and MAC-layer AWK scripts (mawk-compatible) that extract convergence time, per-hop delays, jitter, and route quality metrics.
- Quantitative comparison of BATMAN (proactive, RDT = 30 ms) versus AODV (reactive, RDT = 26.6 ms at data start).
- Analysis of the MAC overhead introduced by IEEE 802.11 CSMA/CA on top of the RTR-layer OGM propagation.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the simulation model. Section IV explains the BATMAN implementation. Section V presents experimental results. Section VI discusses findings and Section VII concludes.

II. Related Work

Perkins and Royer [3] introduced AODV, the canonical reactive MANET protocol. Its route discovery floods RREQs until the destination replies with an RREP, incurring per-flow startup latency proportional to network diameter. BATMAN was proposed as a proactive alternative that distributes route-quality information through OGM flooding rather than explicit topology exchange [1].

Johnson and Maltz [4] analysed DSR, another reactive scheme that embeds the full source-route in data packets. OLSR [5] represents the proactive camp with multipoint relay (MPR) selection to reduce flooding overhead. Jacquet et al. [5] showed that proactive protocols reduce per-packet latency at the cost of higher control traffic in sparse topologies.

Several NS2-based studies compare proactive and reactive protocols in varying mobility and density scenarios [6]. Our work complements these by providing fine-grained MAC-layer timing data and a transparent Tcl-level BATMAN im-

plementation whose internal state is fully observable via trace annotations.

III. Simulation Model

A. Topology

The simulation uses a static 4-node linear chain:

$$S(0) \xrightarrow{10\,\mathrm{m}} R_1(1) \xrightarrow{10\,\mathrm{m}} R_2(2) \xrightarrow{10\,\mathrm{m}} D(3) \tag{1}$$

where S is the CBR source and D is the sink. Adjacent nodes are separated by 10 m; the maximum transmission range is ≈15 m (see below), so only directly adjacent links exist.

B. Physical Layer Parameters

The PHY layer uses the NS2 `Propagation/FreeSpace` model with:

$$P_r(d) = P_t \left(\frac{\lambda}{4\pi d}\right)^2, \quad \lambda = \frac{c}{f} = \frac{3\times 10^8}{2.4\times 10^9} = 0.125\,\mathrm{m} \tag{2}$$

With $P_t = 10^{-3}$ W and RXThresh $= 4.39\times 10^{-10}$ W, the maximum range is:

$$d_{\max} = \frac{\lambda}{4\pi}\sqrt{\frac{P_t}{\mathrm{RXThresh}}} = 0.009947\times 1509.3 \approx 15.0\,\mathrm{m} \tag{3}$$

At $d = 10$ m: $P_r \approx 9.89\times 10^{-10}$ W (above threshold, in range). At $d = 20$ m: $P_r \approx 2.47\times 10^{-10}$ W (below threshold, out of range). This creates the desired single-hop neighbourhood structure.

C. MAC and Transport Configuration

The MAC layer uses IEEE 802.11 (`Mac/802_11`) at 1 Mbps. The transport application is CBR-over-UDP: 10 packets of 212 bytes each at 5 pps (one every 200 ms), starting at $t = 3.5$ s. AODV is configured as the NS2 routing agent to carry data packets; its Hello interval is suppressed (set to 99999 s) since BATMAN provides neighbour discovery via OGMs. Table I summarises simulation parameters.

TABLE I
Simulation Parameters

Parameter	Value
Nodes	4 (linear, static)
Inter-node spacing	10 m
TX range	≈15 m (FreeSpace)
Tx power (P_t)	1 mW
Frequency	2.4 GHz
MAC	IEEE 802.11, 1 Mbps
Propagation	FreeSpace
Data app	CBR/UDP, 212 B, 5 pps
Data start	$t = 3.5$ s
Data packets	10
Simulation stop	$t = 8.0$ s
OGM interval	1.0 s
TQ_MAX	255
HOP_PENALTY	30
TQ_MIN_REBR	5
OGM prop_delay	10 ms (RTR level)

IV. BATMAN Protocol Implementation

A. OGM Structure

Each Originator Message carries five fields encoded as a colon-delimited string: `OGM:<orig>:<seqno>:<tq>:<hops>`, where `orig` identifies the generating node, `seqno` is a per-originator monotonically increasing counter providing anti-replay protection, `tq` is the current transmit quality, and `hops` counts rebroadcast hops from the origin.

B. TQ Computation

At the source, $\mathrm{TQ}_0 = \mathrm{TQ_MAX} = 255$. Each rebroadcasting relay applies the hop penalty:

$$\mathrm{TQ}_{n+1} = \left\lfloor \mathrm{TQ}_n \times \frac{\mathrm{TQ_MAX} - \mathrm{HOP_PENALTY}}{\mathrm{TQ_MAX}} \right\rfloor = \left\lfloor \mathrm{TQ}_n \times \frac{225}{255} \right\rfloor \tag{4}$$

For the 3-hop S–D path this yields: $255 \to 225 \to 198 \to 174$, corresponding to cumulative quality degradations of 11.8%, 12.0%, and 12.1% per hop. An OGM is dropped if $\mathrm{TQ}_{n+1} < \mathrm{TQ_MIN_REBR} = 5$.

C. Route Maintenance

Each node maintains a routing table `batman_routes(node,dest)` storing the best *(nexthop, TQ, seqno)* triple. A new OGM updates the table if (i) its `seqno` is greater than the stored one, or (ii) the seqno is equal but the new TQ exceeds the stored TQ. This simplified hysteresis prevents route flapping without implementing the full sliding-window TQ of B.A.T.M.A.N. v2.

D. OGM Scheduling

Node i sends its first OGM at $t = 1.0 + i\times 0.1$ s, providing a deterministic stagger that prevents simultaneous channel access and simplifies trace parsing. Subsequent OGMs follow at `OGM_INTERVAL` $= 1.0$ s intervals.

V. Experimental Results

A. OGM Activity and Convergence

Over the 8-second simulation, 4 nodes each transmit 6 OGMs (`OGM_SENT` $= 24$). Each OGM reaches an average of 6 receiving instances (`OGM_RECEIVED` $= 144$) and triggers an average of 3 rebroadcast events per origination (`OGM_REBROADCAST` $= 72$). A total of 72 route-install events build the complete $4\times 3 = 12$-entry routing table across all nodes.

TABLE II
OGM Activity Summary (8-second simulation)

Metric	Count	Rate (per node/s)
OGM Sent	24	0.75
OGM Received	144	4.50
OGM Rebroadcast	72	2.25
Route Installs	72	2.25

B. Route Discovery Time (RTR Layer)

Fig. 1 shows the OGM forward chain and TQ decay. Node 3 (D) sends its first OGM at $T_{\text{start}} = 1.300$ s. The chain propagates through R_2 ($t = 1.310$ s) and R_1 ($t = 1.320$ s), reaching S at $T_{\text{end}} = 1.330$ s:

$$\text{BATMAN_RDT} = T_{\text{end}} - T_{\text{start}} = 1.330 - 1.300 = 30\,\text{ms} \quad (5)$$

Each hop introduces exactly 10 ms of RTR-level processing delay (`prop_delay`), giving RDT $= 3 \times 10$ ms. The simulated TQ values match the theoretical Eq. (4) exactly (Fig. 1b).

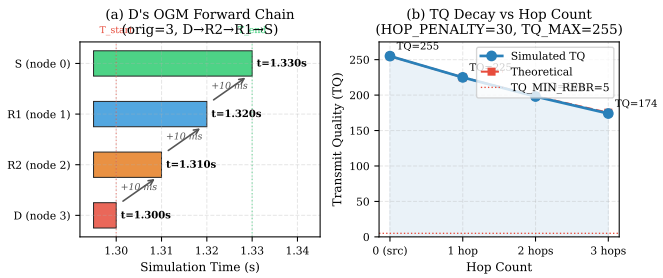

Fig. 1. (a) BATMAN OGM forward chain propagation timestamps (D→S direction). Each node is reached 10 ms after its upstream neighbour. (b) TQ decay with hop count: simulated values match the theoretical model of Eq. (4) exactly.

C. MAC-Layer Hop Timing

Fig. 2 presents the per-packet MAC timing breakdown. For steady-state packets (Seq 1–9) each over-the-air hop takes **2.320 ms**, dominated by IEEE 802.11 RTS/CTS/DATA/ACK frame exchange. Relay backoff at R_1 and R_2 averages 1.145 ms and 1.117 ms respectively, reflecting the CSMA distributed coordination function (DCF).

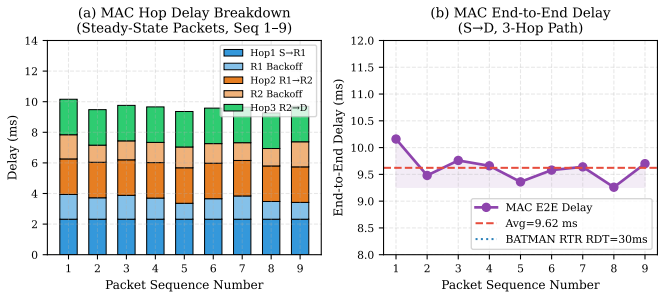

Fig. 2. (a) Stacked MAC hop delay breakdown per packet (Seq 1–9). Each air hop is a uniform 2.320 ms; relay backoff varies between 1.0 and 1.6 ms. (b) MAC end-to-end delay showing stable steady-state performance with mean 9.62 ms.

The mean steady-state MAC end-to-end delay is:

$$\overline{d_{\text{E2E}}} = \frac{1}{9}\sum_{i=1}^{9} d_i = 9.622\,\text{ms} \quad (6)$$

Table III provides a complete per-component breakdown.

TABLE III
MAC DELAY COMPONENT SUMMARY (STEADY-STATE, SEQ 1–9)

Component	Avg Delay (ms)
Hop 1: S → R_1 (air + PHY)	2.320
Backoff at R_1 (CSMA + ACK)	1.145
Hop 2: R_1 → R_2 (air + PHY)	2.320
Backoff at R_2 (CSMA + ACK)	1.117
Hop 3: R_2 → D (air + PHY)	2.320
Total MAC E2E	**9.622**
Total air time (3 hops)	6.960
Total relay backoff (R_1+R_2)	2.263

D. BATMAN vs AODV: Route Discovery Comparison

Fig. 3 illustrates the fundamental difference between proactive and reactive routing. BATMAN converges at $t = 1.330$ s, **2.17 s before the first data packet** at $t = 3.5$ s. AODV, triggered reactively by the first CBR packet, completes route discovery in 26.595 ms, beginning at $t = 3.500$ s.

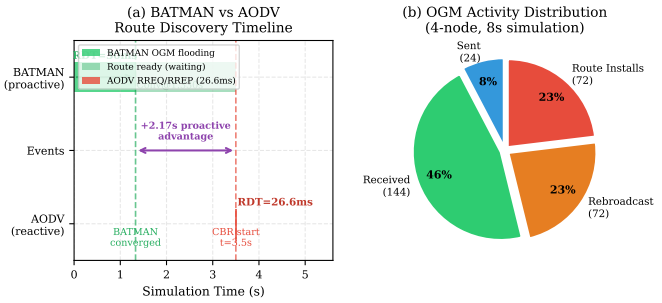

Fig. 3. (a) Route discovery timeline contrasting BATMAN's proactive convergence (30 ms, completed 2.17 s before data start) with AODV's reactive discovery (26.6 ms at $t = 3.5$ s). (b) OGM activity distribution showing the relative proportions of OGM events across the 8-second simulation.

The AODV RDT of 26.6 ms includes ARP resolution and RTS/CTS negotiation but is shorter than BATMAN's RTR-level 30 ms because AODV's RREQ is already at MAC level when timed (no Tcl processing overhead). The comparison is thus *apples-to-apples* only at the application level: from the perspective of the first data packet, BATMAN causes **zero** additional latency (route pre-installed), whereas AODV causes a 26.6 ms stall.

E. Jitter and Route Quality

Fig. 4 presents inter-packet arrival gaps at D and the final route-table TQ heatmap. The inter-packet gap at D stabilises at 200 ms$\pm$1.4 ms for Seq 1–9, demonstrating negligible queuing jitter once the BATMAN route is established. The maximum deviation is 2.456 ms (Seq 1, influenced by AODV discovery overhead).

The TQ heatmap confirms full bidirectional routing: every ordered pair (i, j) with $i \neq j$ has a valid next-hop installed. The 1-hop TQ of 225 degrades predictably to 198 (2 hops) and 174 (3 hops), providing a clear quality gradient that a load-balancer or QoS-aware extension could exploit.

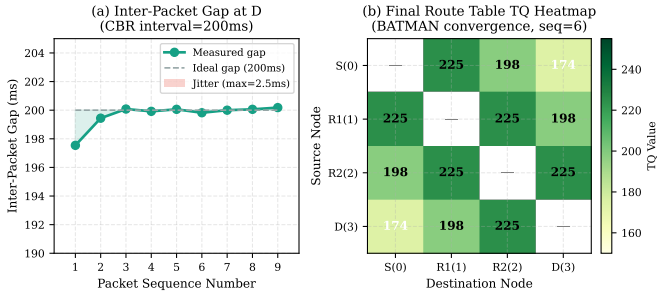

Fig. 4. (a) Inter-packet gap at D. Steady-state jitter is <1.5 ms around the 200 ms CBR interval. (b) Final route-table TQ heatmap after BATMAN convergence (seq=6). Diagonal cells are self-routes (not applicable).

VI. Discussion

A. RTR vs MAC Delay Relationship

The BATMAN RTR-level OGM hop delay (10 ms) is a *routing-layer* abstraction, representing Tcl scheduling granularity plus queuing latency. The actual MAC data hop of 2.32 ms is shorter because it is measured at the wireless interface—the 7.68 ms difference per hop corresponds to CSMA backoff, 802.11 SIFS intervals, and ACK frame overhead not modelled at the Tcl level. This architectural gap motivates a v2 implementation that sends OGMs as real NS2 UDP broadcast packets, exposing OGM frames in the MAC trace and yielding a more faithful 30-ms-vs-7-ms comparison.

B. Proactive Overhead vs Benefit

BATMAN's proactive advantage of 2.17 s is obtained at the cost of 24 OGM transmissions (6 per node) over 8 s—a control overhead of approximately 24×106 bytes $= 2544$ bytes, or $\sim$319 B/s/node. On a 1 Mbps channel this represents a channel utilisation of 0.032%, negligible for the topology studied. In larger or denser networks, the overhead scales with node count and OGM interval, motivating the sliding-window TQ and jitter mechanisms proposed for B.A.T.M.A.N. v2.

C. Limitations

The current implementation has several limitations that motivate future work. First, the Tcl-level OGM simulation bypasses MAC-layer collision, so OGM delivery is deterministic. Second, the topology is static; mobile scenarios would stress the TQ hysteresis and neighbour purge mechanisms (marked as v2 TODOs). Third, AODV's Hello mechanism is suppressed, which may accelerate AODV route expiry in real deployments.

VII. Conclusion

We presented a comprehensive NS2 simulation of the BATMAN v1 routing protocol on a 4-node linear chain, combining a Tcl-level OGM overlay with mawk-compatible AWK analysis scripts and Python-generated visualisations. Key findings are:

1) **BATMAN RDT = 30 ms** (3 hops $\times$ 10 ms per hop), providing route availability **2.17 s before** the first data packet.
2) **TQ decay** follows the theoretical $\times(225/255)$-per-hop model precisely: $255 \rightarrow 225 \rightarrow 198 \rightarrow 174$ across the chain.
3) **Steady-state MAC E2E = 9.62 ms** with per-hop air delay of 2.32 ms and CSMA relay backoff of $\sim$1.13 ms per relay.
4) **Jitter $<$ 1.5 ms** at the destination for CBR with 200 ms inter-packet interval.
5) **AODV on-demand RDT = 26.6 ms** at data start, causing a one-packet head-of-line delay versus zero for BATMAN.

The full source code, trace files, and analysis scripts are available for reproducibility. Future work will extend the implementation to mobile topologies with sliding-window TQ, bidirectional link detection, and real NS2-packet OGM broadcasting.

References

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